

C&I ESS CABINET

ESS-TRENE

Air cooling



PARTNER



SAFE

- Intelligent air cooling for optimal heat dissipation
- Four-level fire protection
- AC&DC type II SPD



ECONOMIC

- Advanced LFP battery quality assured
- High power density with less space
- Expandable to MWh



INTELLIGENT

- Smart energy management strategy*
- VPP ready, SolaX cloud supports resource aggregator (2030.5, OpenADR)
- Support Micro-grid and a variety of scenarios
- Support 7x24h remote O&M and schedule deployment



ROBUST

- Support on-grid and off-grid solution
- Self-developed BMS&EMS on SolaX Cloud platform
- 1 year history system data backed up on EMS
- Support Cell level balance, smart temperature check and control

* Under development



TRENE series C&I energy storage cabinet

is a highly integrated, all-in-one solution with versatile application scenarios.

TRENE series C&I energy storage cabinet is a highly integrated, all-in-one solution with versatile application scenarios. TRENE air-cooled series provides efficient, safe, and stable smart energy storage solutions.

Firstly, the cabinet adopts high-density, high-safety, and high-performance LFP cells. With a capacity of 215kWh per cabinet, it can reliably perform charging and discharging operations for single or multiple cabinets, with a lifespan of over 10 years. The large-capacity 280Ah battery cells also reduce the overall system investment cost.

Secondly, the cabinet is equipped with a self-developed Energy Management System (EMS) that can monitor the working status and abnormal alerts of each battery cell, PCS, and fire protection system in real-time. The local data storage capability allows for data analysis and verification for up to 1 year. The advanced EMS system also has leading advantages in intelligent control of different smart operation strategies,

TRENE “INTRODUCTION”

autonomous scheduling based on local electricity prices, and comprehensive management of photovoltaic, energy storage systems, EV charging and generators at power plant level. These features improve the overall system efficiency and shorten the investment return period.

Additionally, the cabinet integrates multiple safety protection measures. It has built-in protection functions such as overvoltage, overcurrent, and over-temperature, as well as fire-resistant materials and 4-level fire protection system to promptly detect and respond to potential fire risks. This effectively controls the spread of fires and reduces the risk of safety accidents.

The cabinet is suitable for various commercial and industrial scenarios, including peak shaving, demand response, backup mode, photovoltaic and energy storage integration, and stable load consumption curves. It also supports applications such as virtual power plants (VPP) and frequency regulation.

TRENE-P100B215I

AC Side	
Rated AC power [kW]	100
Rated AC current [A]	144.4
Max. AC apparent power [kVA]	110
Nominal grid voltage [V]	400 (-20% ~ +15%)
Nominal grid frequency [Hz]	50 / 60
Adjustable power factor range	0.99leading ~ 0.99lagging
THDi (Rated power) [%]	< 3
Max. efficiency [%]	98%
Battery	
Battery type	LFP 280Ah
Battery capacity [kWh]	215
Rated battery voltage [V]	768
Battery voltage range [V]	600 ~ 876
Discharge depth [%]	90
Rated charge/discharge current [A]	140
General	
Dimension (W×H×D)[mm]	1680 × 2420 × 1200
Weight [kg]	2800
Operating temprature range [℃]	-30 ~ 55
Relative humidity(Non-condensing) [%]	0 ~ 95
Altitude [m]	3000
Cooling concept	Smart air cooling
Ingress protection	IP55
Fire protection	Aerosol(Optional:Novec1230) / Water
Topology	Non-isolated
Certificates	IEC62619, IEC63056:2000, IEC61000, IEC62477-1, UN38.3, GB/T36276, GB/T34131